

Fraud Detection Dataset - Insights Report

Data Overview

Data contains 30 features: 28 anonymous (PCA), 2 (Time, Amount), and 1 target (Class).

Dataset size: 170,884 samples.

Data types: float64 (30 features), int64 (target).

No null values detected.

Class Imbalance

Mean of Class = 0.001785 → severe imbalance.

Data Cleaning

448 duplicated transactions (0.26%) → dropped.

Metrics Note

Accuracy is misleading due to imbalance.

Feature Engineering

Converted Time (seconds) to Hours (0–23).

Time-Based Insights

Normal transactions peak from 9:00 to 22:00.

Fraud spikes at 02:00 AM (low surveillance) and 11:00 AM (camouflage).

Amount Insights

Right-skewed distribution (mean > median).

Median = 16.48 → micro-transactions preferred.

Max = 1504 → strong contrast.

Amount is not a strong feature overall.

Feature Importance

Negative region importance: V14, V12, V10, V17.

Positive region importance: V4, V11.

Separated features: V3, V9, V16.

Some features are not informative.